

MODULE 3 – DISPERSION

Mean Deviation + Standard Deviation: Master Decision Sheet

Individual, Discrete and Continuous Series

1. The Master Idea

Do not memorize nine different methods. The common pattern is:

Identify data type → Find center → Find deviations → Apply formula

Data	Given	Main extra step
Individual	Values only	No frequency; effectively $f = 1$.
Discrete	X, f	Use frequency; use cf when locating median/quartiles.
Continuous	Class intervals, f	Use midpoint for mean/MD/SD; use grouped formulas for median/mode.

2. Mean Deviation (MD)

Mean Deviation is the average distance of observations from a chosen center. The center can be mean $\bar{X}$, median M , or mode Z .

$$MD = \frac{\sum f|X - A|}{N}$$

For individual data, $f = 1$:

$$MD = \frac{\sum |X - A|}{N}$$

where A is the chosen center.

Absolute value: $|X - A|$ means distance only; ignore the negative sign.

CENTER → DISTANCE → WEIGHT BY f → ADD → DIVIDE

Coefficients of MD

$$C_{\text{MD, mean}} = \frac{MD}{\bar{X}}$$

$$C_{\text{MD, median}} = \frac{MD}{M}$$

$$C_{\text{MD, mode}} = \frac{MD}{Z}$$

3. MD: Three Data Arrangements

Individual Series

No frequency is given. Effectively $f = 1$.

$$\bar{X} = \frac{\sum X}{N}$$

About mean:

$$MD = \frac{\sum |X - \bar{X}|}{N}$$

About median:

$$MD = \frac{\sum |X - M|}{N}$$

About mode:

$$MD = \frac{\sum |X - Z|}{N}$$

Discrete Series

First:

$$N = \sum f \quad \bar{X} = \frac{\sum fX}{N}$$

Then calculate $|X - A|$, $f|X - A|$, and:

$$MD = \frac{\sum f|X - A|}{N}$$

Useful table:

X ,	f ,	fX ,	$ X - A $,	$f X - A $
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Continuous Series

First convert each class to its midpoint:

$$X = \frac{\text{lower limit} + \text{upper limit}}{2}$$

Then use the same frequency MD formula:

$$MD = \frac{\sum f|X - A|}{N}$$

For median/mode, first find the grouped-data center, then use it as A .

4. Standard Deviation (SD)

SD measures spread around the mean. Deviations are squared, averaged, then square-rooted.

$$\sigma = \sqrt{\frac{\sum (X - \bar{X})^2}{N}}$$

For frequency data:

$$\sigma = \sqrt{\frac{\sum f(X - \bar{X})^2}{N}}$$

$$\text{MEAN} \rightarrow \text{DEVIATION} \rightarrow \text{SQUARE} \rightarrow \text{AVERAGE} \rightarrow \sqrt{}$$

Core difference:

$$\text{MD uses } |X - A| \quad \quad \text{SD uses } (X - \bar{X})^2$$

5. SD: Individual Series

Find:

$$\bar{X} = \frac{\sum X}{N}$$

Then:

$$d = X - \bar{X}, \quad d^2 = (X - \bar{X})^2$$

and:

$$\sigma = \sqrt{\frac{\sum d^2}{N}}$$

Shortcut:

$$\sigma = \sqrt{\frac{\sum X^2}{N} - \bar{X}^2}$$

6. SD: Discrete Series

Given X, f :

$$N = \sum f \quad \bar{X} = \frac{\sum fX}{N}$$

Direct method:

$$d = X - \bar{X}, \quad fd^2 = f(X - \bar{X})^2$$

$$\sigma = \sqrt{\frac{\sum f(X - \bar{X})^2}{N}}$$

Shortcut:

$$\sigma = \sqrt{\frac{\sum fX^2}{N} - \bar{X}^2}$$

7. SD: Continuous Series

For every class, find the midpoint:

$$X = \frac{L + U}{2}$$

Then treat the midpoint as X and use:

$$\sigma = \sqrt{\frac{\sum fX^2}{N} - \bar{X}^2}$$

with:

$$\bar{X} = \frac{\sum fX}{N}$$

Thus:

$$\text{Class interval} \rightarrow \text{midpoint } X \rightarrow fX, fX^2 \rightarrow \bar{X} \rightarrow SD$$

8. Assumed Mean / Step-Deviation

If taught by your lecturer, let:

$$d = X - A$$

Then:

$$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2}$$

For equal class width i :

$$u = \frac{X - A}{i}$$

$$\sigma = i \sqrt{\frac{\sum fu^2}{N} - \left(\frac{\sum fu}{N}\right)^2}$$

Only use these if this method is required in your class.

9. Quick Comparison

	Mean Deviation	Standard Deviation
Basic idea	Average absolute distance	Square-root of average squared distance
Deviation	$ X - A $	$(X - \bar{X})^2$
Frequency formula	$\frac{\sum f X - A }{N}$	$\sqrt{\frac{\sum f(X - \bar{X})^2}{N}}$
Key operation	Absolute value	Square, average, square root

10. Formula Decision Sheet

Range:

$$R = L - S$$

Coefficient of Range:

$$\frac{L - S}{L + S}$$

Interquartile Range:

$$IQR = Q_3 - Q_1$$

Quartile Deviation:

$$QD = \frac{Q_3 - Q_1}{2}$$

Coefficient of QD:

$$\frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Mean Deviation:

$$MD = \frac{\sum f|X - A|}{N}$$

Ask: “About what?” Then use $A = \bar{X}$, M , or Z .

Standard Deviation:

$$\sigma = \sqrt{\frac{\sum f(X - \bar{X})^2}{N}}$$

Shortcut:

$$\sigma = \sqrt{\frac{\sum fX^2}{N} - \bar{X}^2}$$

For individual data, take $f = 1$.

For continuous data:

$$X = \frac{\text{lower limit} + \text{upper limit}}{2}$$

11. Exam Panic Protocol

When a huge table appears, ask only:

1. What type of data is this?

2. What center/formula do they want?

3. What column do I need next?